

Programming Using C++ Level 1

Variables



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Variables

Definition

- ◆ A variable is a named storage location in memory used to store a value that can be used and changed while the program is running.
- ◆ Every variable has a data type, a name, and a value.

Explanation

■ In C++, a variable is commonly created using this structure:

```
string Name = "Sara Alterkawi";
```

■ string → tells C++ that the variable stores text.

■ Name → the name of the variable.

■ "Sara Alterkawi" → the value assigned to the variable.

◆ string is used to store text, such as names, cities, and countries.

```
string Name = "Sara Alterkawi";
```

◆ int is used to store ****whole numbers**** without decimal places.

```
int Age = 43;
```

■ The value is 43.

■ It does not contain a decimal part.

◆ float is used to store numbers that can contain decimal values.

```
float MonthlySalary = 40000;
```

◆ A float can also store values such as:

```
float Price = 19.95;
```

◆ char is used to store a single character.

```
char Gender = 'F';
```

◆ bool is used to store a Boolean value, which can be either:

▨ true or false

```
bool isMarried = true;
```

◆ Here, isMarried stores the Boolean value true.

Steps in main()

1. Declare the variables

- ◆ Create variables and assign values to them.

2. Store the data

- ◆ Each variable stores a specific type of value.

3. Perform calculations

- ◆ Use stored values to calculate new values, such as yearly salary.

4. Display the information

- ◆ Use cout to print the values on the screen.

5. End the program

- ◆ Use return 0; to indicate that the program finished successfully.

💡 Example

```
cout << "Name: " << Name << endl;
```

■ "Name: " → text displayed on the screen.

■ Name → the value stored in the variable.

■ endl → moves the cursor to a new line.

Note

◆ The = operator here is called the assignment operator. It assigns the value on the right to the variable on the left.

◆ A char value is written using single quotes ' '.

◆ "F" would be a string, while 'F' is a character.

◆ When you write:

```
cout << isMarried;
```

◆ C++ normally displays:

1 for true, and:

0 for false.

💡 If you want C++ to display the words `true` and `false`, you can use:

```
cout << boolalpha << isMarried;
```

```
true
```

⊘ Warning

- ◆ Do not use double quotes for a char:
char Gender = "F"; // ✗ Wrong
- ◆ Use single quotes:
char Gender = 'F'; // ✓ Correct
- ◆ For a string, use double quotes:
string Gender = "Female"; // ✓ Correct



Comparison

string

- ✓ Stores text.
- ✓ Example: "Sweden"
- ✓ Uses double quotes.

char

- ✓ Stores one character.
- ✓ Example: 'F'
- ✓ Uses single quotes.

int

- ✓ Stores whole numbers.
- ✓ Example: 43
- ✓ No decimal part.

float

- ✓ Stores floating-point numbers.
- ✓ Example: 19.95
- ✓ Can contain a decimal part.

✦✦ Summary

✓ A variable stores a value that can be used later in the program.

- ▣ string → text

- ▣ int → whole numbers

- ▣ float → decimal/floating-point numbers

- ▣ char → one character

- ▣ bool → true or false

✓ Variables make programs more useful because we can store data, use it in calculations, and display it.

```
data_type variable_name = value;
```

```
int Age = 43;
```

Thank you 😊